P1DF7-F1、P1DF7-F2、P1DF7-F3、P1DF8-F1、P1DF8-F2、P1DF9-F1、P1DF9-F2、P1DFE-F3、P1DFF-F3、P1E00-F2

Fault diagnosis code
P1DF7-F1：Discharge overcurrent level 2 fault
P1DF7-F2：discharge overcurrent3level failure
P1DF7-F3：PackOvercurrent Protection
P1DF8-F1：Charge overcurrent2level failure
P1DF8-F2：Charge overcurrent3level failure
P1DF9-F1：Recharge overcurrent2level failure
P1DF9-F2：Recharge overcurrent3level failure
P1DFE-F3：Single overcharge safety protection
P1DFF-F3：Single over-discharge safety protection
P1E00-F2：Monomer pressure difference is large3level failure
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• BMS fault • BMS insulation fault • BMS high voltage circuit fault • The current circuit of BMS fault
Conditions for fault code operation
Turn the ignition switch to ON position.
Conditions for fault code setting
P1DF7-F1
Greater than the discharge current limit parameter +15A (3A before modification)
P1DF7-F2
Greater than the discharge current limit parameter +15A (3A before modification)
P1DF7-F3
The current is greater than the current limit parameter +20A for 30s (including discharge overcurrent, feedback overcurrent, and charging overcurrent)
P1DF8-F1
Greater than the charging current limit parameter +15A
P1DF8-F2
current continues30sgreater than the charge current limit parameter+15A(3A before modification)
P1DF9-F1
current continues15sGreater than the feedback current limit parameter+15A(3A before modification)

P1DF9-F2
current continues30sGreater than the feedback current limit parameter+15A(3A before modification)
P1DFE-F3
The charging voltage of any cell is higher than the safety threshold:> 3.8V
P1DFF-F3
The charging voltage of any cell is lower than the safety threshold：< 2.0V
P1E00-F2
The battery voltage difference between the maximum battery voltage and the minimum battery voltage is higher than the safety threshold：> 400mV
To set the effect of a fault code condition

Troubleshooting steps	
Step 1	General inspection.
	1 Check battery capacity. 2 Check the harness plug of BMS for damage, poor contact, aging and looseness. 3 Check the harness plug of battery pack high voltage for damage, poor contact, aging and looseness. 4 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 2	Check fault codes.
	1 Connect the fault diagnosis instrument to the diagnosis interface. 2 Turn the ignition switch to ON position. 3 Diagnose the BMS with diagnostic apparatus. 4 If there is any fault code except for P1DF7-F1、P1DF7-F2、P1DF7-F3、P1DF8-F1、P1DF8-F2、P1DF9-F1、P1DF9-F2、P1DFE-F3、P1DFF-F3、P1E00-F2?
	YES → Refer to:DTC Summary list(BMS).
NO ↓	
Step 3	Check the connecting cable of BMS.
	1 Turn the ignition switch to OFF. 2 Make protective preparations according to the safety operation specifications of the high voltage voltage in the maintenance manual. 3 Disconnect the harness plug battery pack high voltage. 4 Disconnect the high voltage harness plug of BMS. 5 Check whether the BMS connection cable is rotten, damaged, and water enters. 6 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 4	Check the insulation of BMS.
	1 Use a megohm meter to check the insulation resistance of the BMS housing and grounding according to the procedure in the maintenance manual. **Standard value （insulation level 1) ＞ 1000Ω/V** **Standard value （insulation level 2) ＞ 600Ω/V** 2 Check whether the result is normal or not?

	NO → Repair or replace the faulted parts.
YES ↓	
Step 5	Check the insulation of BMS high-voltage cables.
	1 Use a megohm meter to check the insulation resistance of the BMS high voltage and grounding according to the procedure in the maintenance manual. **Standard value （insulation level 1) ＞ 1000Ω/V** **Standard value （insulation level 2) ＞ 600Ω/V** 2 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 6	Perform initialization.
	1 Turn the ignition switch to OFF. 2 Connect the fault diagnosis instrument to the diagnosis interface. 3 Turn the ignition switch to ON. 4 The BMS module was initialized with the diagnostic instrument. 5 Can the BMS be initialized and is the BMS restored?
	NO → Repair or change the BMS module.
YES ↓	
Step 7	Test whether BMS is norm.
	1 Turn the ignition switch to OFF. 2 Disconnect and check the harness plug CH1001 of BMS. 3 Check and repair. • Corrosion(install new adapting piece or terminal-cleaning module and pin). • Pin damaged or bent-install new terminal/pin. • Fly out of pins-install new pins as needed. 4 Reconnect the BMS plug and all other previously disconnected plug. Make sure the pin is properly positioned and fixed. 5 Run the system and determine if the problem persists?
	YES → Replace the BMS.
NO ↓	
At this point the system is functioning normally. The problem may be caused by module connections. Address the root cause of any connector or pin problems.	
Next Step ↓	

Diagnosis end.